



Load-aware Ground Station Assignment for Low Earth Orbit Satellite Networks

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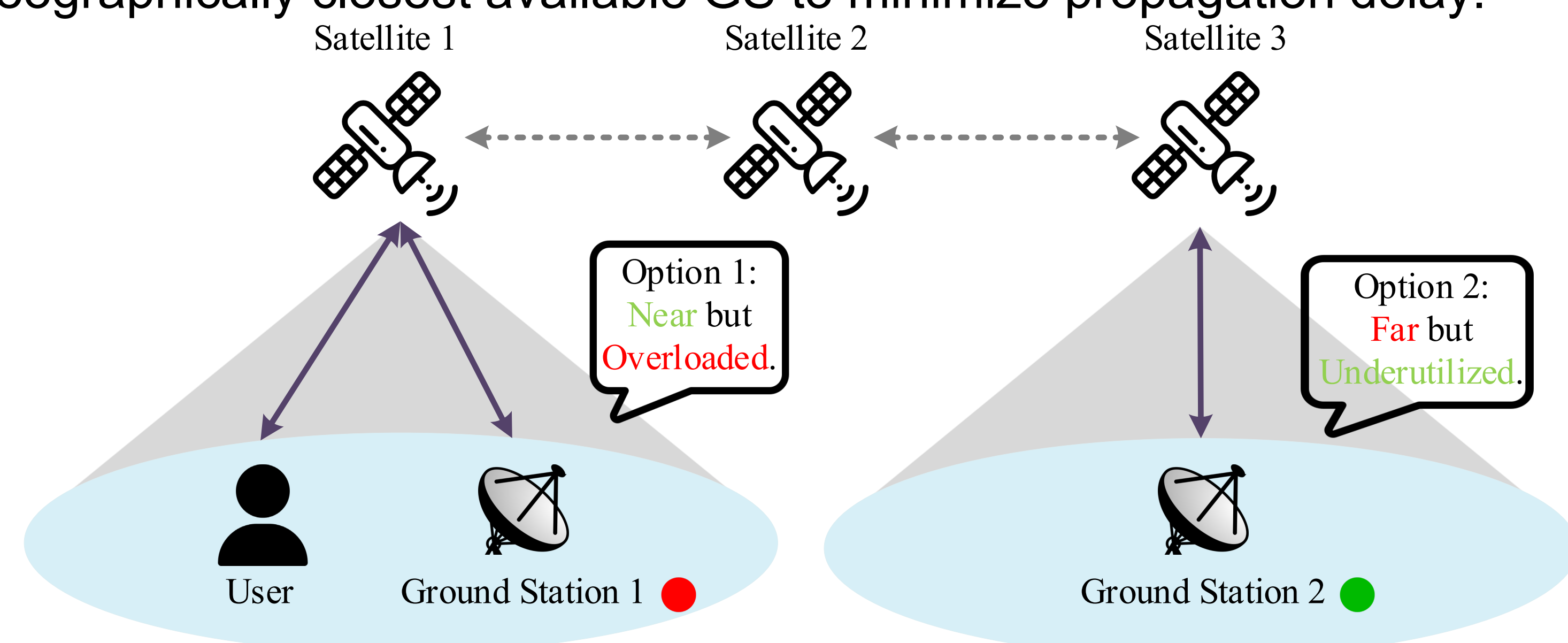
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① Introduction

• The Rise of LEO Mega-Constellations

- **Global Coverage:** LEO constellations like Space's Starlink and OneWeb are deploying thousands of satellites to reshape global broadband.
- **"Bent-pipe" Model:** Traditional designs rely on a user terminal connecting to an overhead satellite, which immediately relays data to a local Ground Station (GS).
- **Dependency:** This architecture relies heavily on a dense network of GSs to connect users to the terrestrial Internet.
- **Current Routing Standard:** The industry-standard "Nearest-GS" strategy assigns users to the geographically closest available GS to minimize propagation delay.



An illustrative example.

② Motivation & Challenges

• The Ground Segment Bottleneck

- **Uneven Traffic Demand:** User traffic is not spatially uniform; dense populations generate heavy loads while oceans/rural areas have little usage, creating "hotspots".
- **Capacity Constraints:**
 - Each GS has a fixed aggregate Ground-to-Satellite Link (GSL) capacity.
 - When high user demand is funneled to a single "nearest" GS, this capacity is quickly saturated.
- **The Consequence:** The nearest-GS approach leads to severe load imbalance, where specific GSs become overloaded (utilization >100%), causing packet loss and service degradation.

• The Opportunity: ISL-Enabled Flexibility

- **Dynamic Mesh in Space:** Modern satellites (e.g., Starlink v1.5+) feature Inter-Satellite Links (ISLs), allowing data to hop between satellites before reaching the ground.
- **Offloading Potential:** Traffic can be routed to distant, underutilized GSs to bypass local congestion.
- **Key Challenge:** Offloading increases latency. We must balance *load distribution* against *latency penalties*.

③ Solution – Load-aware Assignment

• Optimization Framework (MILP)

- **Objective:** We formulate a Mixed-Integer Linear Program (MILP) to minimize the peak GSL utilization across all GSs in the network at every time interval.
- **Centralized Control:** The algorithm operates within a centralized controller that updates routing decisions periodically (e.g., every time slot)

• Critical Constraints

- **GS Assignment Constraint:**
 - A user can be served by a GS only if a feasible communication path exists.
- **User Request Constraint:**
 - Each user must be connected to exactly one GS in each time interval to ensure service continuity.
- **User Latency Constraint:**
 - To maintain service quality, the assignment must respect a latency threshold.
 - This ensures the user-experienced delay does not exceed the baseline delay to their nearest GS by more than this factor, preventing unacceptable latency penalties.
- **GSL Utilization Constraint:**
 - The total traffic demand from all users assigned to a particular GS cannot exceed its total downlink capacity.

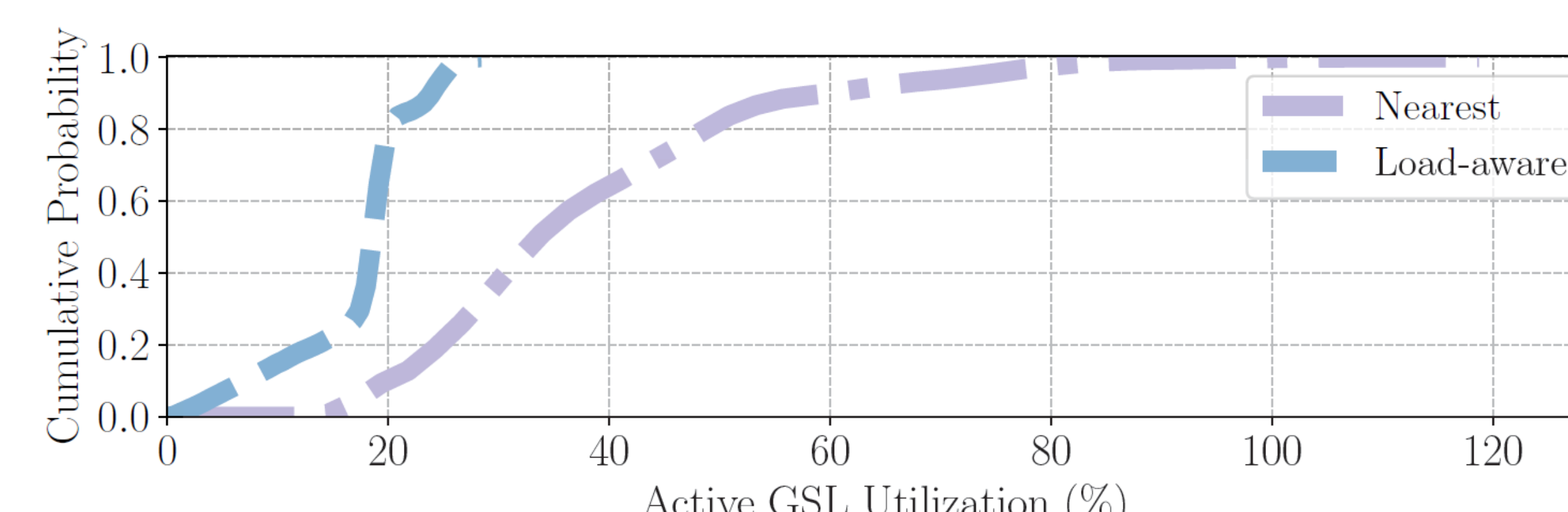
④ Preliminary Results

• Simulation Parameters

- **Constellation:** Modeled after Starlink Phase 1 with 1,584 satellites at 550 km altitude.
- **Ground Segment:** 200+ Ground Stations, each with a GSL capacity of 4 Gbps.
- **Traffic Model:** Demand distribution based on real-world population density.

• Performance Analysis

- **Baseline (Nearest-GS):**
 - Shows a "long tail" distribution in GSL utilization.
 - A significant portion of GSs are severely overloaded, with utilization reaching nearly **120%**.
- **Proposed (Load-aware):**
 - Successfully balances traffic across the network.
 - The CDF curve is tightly clustered, with all active GSL utilization maintained **below 30%**.



CDF of GSL utilization for all active GSs across time.